

Module - II

Groups, Rings, Fields & Lattice

Algebraic Structure (Groupoid)

It is also called groupoid ~~an~~ or quasi-group.

Let S be a non-empty set, then it is called algebraic structure if $(a * b) \in S \quad \forall a, b \in S$.

Closure Property

The groupoid is set ' S ' with a binary operation ' $*$ ' defined on S such that S is closed under the operation ' $*$ ' i.e. ~~$\forall$~~ $a, b \in S, (a * b) \in S$.

$(\mathbb{N}, +) \quad 1+2 \in \mathbb{N}, \quad 3 \in \mathbb{N} \quad [\text{Groupoid}]$

Note: Group theory have found wide application in physical sciences and biological science.

Associative Property

A set ' A ' w.r.to operator ' $*$ ' is said to satisfy associative property if $\forall a, b, c \in A$
 $\Rightarrow a * (b * c) = (a * b) * c$.

Semi Group

If any algebraic structure satisfies the associative property then it is called semi-group.

$$(N, +) \quad (2+3)+5 = 2+(3+5)$$

$$10 = 10 \in N \quad [\text{It is semi-group}]$$

Identity Property

A set 'A' w.r.to operator '*' is said to satisfy identity property if $\forall a \in A$ such that there exists an element 'e' i.e. $a * e = a = e * a$.

$$a + 0 = a \Rightarrow a + e = a$$

$$a \times 1 = a \Rightarrow a \times e = a$$

Monoid

If a semi-group satisfies identity property, then it is called monoid.

$$(N, \times) \quad 5 \times 1 = 5$$

$$5 \in N$$

$$1 \in N$$

It is monoid.

Inverse Property

A set 'A' w.r.to operator '*' is said to satisfy inverse property if $\forall a \in A$ there exists an element a^{-1} such that,

$$a * a^{-1} = a^{-1} * a = e$$

& $a^{-1} \in A$.

for +, $a + (-a) = e$ i.e. 0

for $\times$, $a \times 1/a = e$ i.e. 1

Imp Group

If a monoid satisfies inverse property then it is called group. A non-empty set 'G' $(G, *)$ is called a group if it follows the following properties:

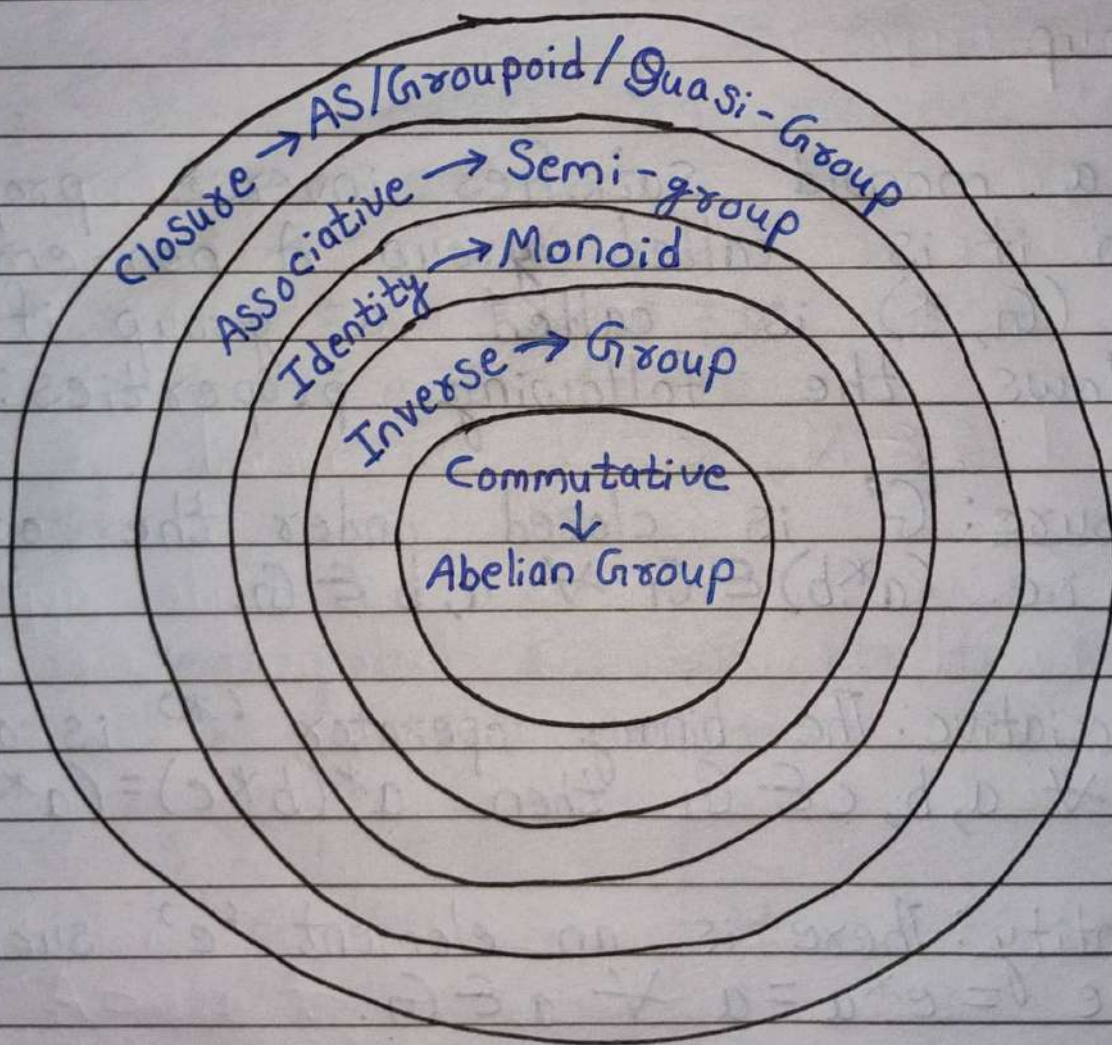
- i. Closure: 'G' is closed under the operation '*' i.e. $(a * b) \in G \forall a, b \in G$.
- ii. Associative: The binary operator '*' is associative if $\forall a, b, c \in G$ then $a * (b * c) = (a * b) * c$.
- iii. Identity: There is an element 'e' such that $a * e = e * a = a \forall a \in G$.
- iv. Inverse: For every element in G, there exists an inverse element in G w.r.to operator '*' i.e. $\forall a \in G, a^{-1} \in G$ such that $a * a^{-1} = a^{-1} * a = e$.

Abelian Group

A set ' G ' with operator ' $*$ ' is said to satisfy commutative property if $\forall a, b \in G$ such that $a * b = b * a$.

If a group satisfies commutative property then it is called Abelian group.

Hierarchy of Algebraic Structures (A.S.)



Q. Prove that a group G is abelian if and only if $(ab)^2 = a^2b^2 \forall a, b \in G$.

Solⁿ:

Given, $(ab)^2 = a^2b^2$

Expanding both sides,

$$(ab)(ab) = (aa)(bb)$$

Applying associative property,

$$a(ba)b = a(ab)b$$

$$\Rightarrow ba = ab$$

Hence, G is an abelian group.

9. Check that $[1, 0, X]$ is group or not?

Solⁿ:

X	1	0
1	1	0
0	0	0

← Composition Table
or Cayley Table

① For closure property, $(a \times b) \in G$. 0 & 1 belongs to group. Therefore, it satisfies the closure property, it is algebraic structure.

② For associative property, $a \times (b \times c) = (a \times b) \times c$. $0 \in G$, it satisfies the associative property, so it is semi-group.

③ For identity property, $a \times e = e \times a = a$

$$a \times 1 = 1 \times a. \quad [\text{let } e = 1]$$

$$1 \times 1 = 1 \checkmark \quad \& \quad 0 \times 1 = 0 \times$$

It ^{does not} satisfies the identity property, so it is ^{not} monoid.

Not Necessary to check inverse if it is not monoid.

④ For inverse property, $a \times a^{-1} = e = 1$ & $a^{-1} \in G$

$$\text{Let, } a = 1 \Rightarrow 1 \times 1^{-1} = e = 1. \quad \left[\begin{array}{l} X, e = 1 \\ +, e = 0 \end{array} \right]$$

For 0, it does not satisfies the inverse property.

Therefore, it is not group.

Q. Check $[1, -1, \times]$ is group or not?

Solⁿ:

X	1	-1
1	1	-1
-1	-1	1

} Composition Table
(or Cayley)

For closure property, $(1 \times -1) \in G$, it is algebraic structure.

For associative property, $a \times (b \times c) = (a \times b) \times c$.

Let, $a = 1, b = 1, c = -1$

$$1 \times (1 \times -1) = (1 \times 1) \times -1$$

$$\Rightarrow -1 = -1$$

$$\Rightarrow 1 = 1$$

It satisfies the associative property, so it is semi-group.

For identity property, $a \times e = e \times a$.

$$a \times 1 = 1 \times a$$

$$\text{Let, } a = 1, \Rightarrow 1 = 1$$

$$\text{Let, } a = -1 \Rightarrow 1 = 1$$

It satisfies the identity property, so it is monoid.

For inverse property, $a \times a^{-1} = e = 1$ & $a^{-1} \in G$.

$$\text{Let, } a = 1 \Rightarrow 1 \times \frac{1}{1} = 1 = e$$

$$\text{Let, } a = -1 \Rightarrow -1 \times \frac{1}{-1} = 1 = e$$

It satisfies the inverse property.

Thus, it is group.

Q. Check $[1, -1, i, -i], \times$, $i^2 = -1$ is group or not?

Solⁿ:

Composition table,

$\times$	1	-1	i	-i
1	1	-1	i	-i
-1	-1	1	-i	i
i	i	-i	$i^2(-1)$	$-i^2(1)$
-i	-i	i	$-i^2(1)$	$i^2(-1)$

For closure property, $(a \times b) \in G$, $1, -1, i, -i$ belongs to group.

It satisfies the closure property, so it is algebraic structure.

For associative property, $a \times (b \times c) = (a \times b) \times c$

Let, $a = 1$, $b = -1$ & $c = i$

$$1 \times (-1 \times i) = (1 \times -1) \times i$$

$$\Rightarrow i = i$$

It satisfies the associative property, so it is semi-group.

For identity property, $a \times e = e \times a = a$

$$a \times 1 = 1 \times a = a$$

Here $e = 1 \in G$

^ It satisfies the identity property, so it is monoid.

For inverse property, $a \times a^{-1} = a^{-1} \times a = e$
& $a^{-1} \in G$.

$$1 \times \frac{1}{1} = 1 = e$$

$$-1 \times \frac{1}{-1} = 1 = e$$

$$i \times \frac{1}{i} = 1 = e$$

$$-i \times \frac{1}{-i} = 1 = e$$

It satisfies, so it is group.

Q. Check $(\{1, \omega, \omega^2, X\})$ is group or not, given that $\omega^3 = 1$?

Solⁿ:

Composition table,

X	1	ω	ω^2
1	1	ω	ω^2
ω	ω	ω^2	1
ω^2	ω^2	1	ω

For closure property, $(a \times b) \in G$ i.e. 1, ω & ω^2 belongs to G . It is algebraic structure.

For associative property, $a \times (b \times c) = (a \times b) \times c$

$$1 \times (\omega \times \omega^2) = (1 \times \omega) \times \omega^2 \rightarrow \left. \begin{array}{l} \text{For } a=1 \\ b=\omega \\ c=\omega^2 \end{array} \right\}$$
$$\Rightarrow \omega^3 = \omega^3$$

It is semi-group.

For identity property, $a \times e = e \times a = a$
 $\Rightarrow a \times 1 = 1 \times a = a$

$$e = 1 \in G$$

It is a monoid.

For inverse property, $a \times a^{-1} = a^{-1} \times a = e = 1$ & $a^{-1} \in G$

$$\begin{aligned} (1 \times 1^{-1} = 1) &\in G \\ (\omega \times \omega^{-1} = \omega \times \omega^2 = 1) &\in G \\ (\omega^2 \times \omega^{-1} = \omega^2 \times \omega = 1) &\in G \end{aligned}$$

Hence, it is a group.

$$a +_n b = \text{Remainder of } \frac{a+b}{n}$$

$$OR = (a+b) - n$$

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Addition Modulo

Q. $M((\{0, 1, 2, 3\}, +_4))$, check for group or not?
 $\uparrow$ Modulo $\downarrow$ Base

Solⁿ:

Composition table,

$+_4$	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

If addition makes number greater than the members in the group then we subtract the number from the base, regardless of sign.

For closure property, ~~$(a + b) \in G$~~ $(a +_4 b) \in G$.

In the above table entries belongs to G . So, it is algebraic structure.

For associative property, $a +_4 (b +_4 c) = (a +_4 b) +_4 c$
 $\forall a, b, c \in G$.

$$\text{LHS, } 1 +_4 (2 +_4 3) = 1 +_4 1 = 2$$

$$\text{RHS, } (1 +_4 2) +_4 3 = 3 +_4 3 = 2.$$

It satisfies associative property. So, it is a semi-group.

For identity property, ~~$a \times e = e \times a$~~
 $a +_4 e = e +_4 a = a \quad \forall a \in G$

For $+$, $e = 0 \in G$. 0 is identity element.

It satisfies identity property. So, it is a monoid.

For inverse property, $a \times a^{-1} = a^{-1} \times a = e = 0$
& $a^{-1} \in G$.

$$0^{-1} = 0 \in G \quad [0 +_4 0 = 0]$$

$$1^{-1} = 3 \in G \quad [1 +_4 3 = 0]$$

$$2^{-1} = 2 \in G \quad [2 +_4 2 = 0]$$

$$3^{-1} = 1 \in G \quad [3 +_4 1 = 0]$$

It satisfies inverse property.

Hence, it is a group- (also an abelian group).

Note: For $\times$, $a^{-1} = 1/a$

For $+$, $a^{-1} = -a$ but only when no base
but for base 4, a^{-1} is that what
number added to a gives 0 under
base/modulo 4.

Q. Check $(\{0, 2, 3, 4, 6, 8\}, +_9)$ is a group or not?

Solⁿ:

$$G = \{0, 2, 3, 4, 6, 8\}$$

Composition table,

$+_9$	0	2	3	4	6	8
0	0	2	3	4	6	8
2	2	4	5	6	8	1
3	3	5	6	7	0	2
4	4	6	7	8	1	3
6	6	8	0	1	3	5
8	8	1	2	3	5	7

For closure property, $(a +_9 b) \notin G$.

In the above composition table, ^{not} all the entries belong to G - i.e. $(a +_9 b) \notin G \nexists (a, b) \in G$.

It does not satisfy the closure property.

Hence, it is not a group.

Q. Show that the matrix $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$,
 $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ & $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$ form a group w.r.to multiplication?

Solⁿ:

$$\text{Let, } A = \begin{bmatrix} 1 & 0 \\ 0 & +1 \end{bmatrix}, \quad B = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad D = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

Now,

$$A \times A = \begin{bmatrix} 1 & 0 \\ 0 & +1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 \\ 0 & +1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = A$$

$$A \times B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = B$$

$$B \times B = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = A$$

$$C \times C = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = A$$

$$D \times D = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \times \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = A$$

$$A \times C = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = C$$

$$A \times D = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = D$$

$$B \times C = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = D$$

$$B \times D = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = C$$

$$C \times D = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \times \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = B$$

Composition table,

X	A	B	C	D
A	A	B	C	D
B	B	A	D	C
C	C	D	A	B
D	D	C	B	A

For closure property, $(a \times b) \in G$ i.e. A, B, C & D belongs to G .

It is algebraic structure.

For associative property, $a \times (b \times c) = (a \times b) \times c$

Let, $a = A$, $b = B$ & $c = C$

$$A \times (B \times C) = (A \times B) \times C$$

$$\Rightarrow A \times D = B \times C$$

$$\Rightarrow D = D.$$

It is semi-group.

For identity property, $a \times e = e \times a = a$

Here, for X , $e = I$

$$a \times I = a$$

$$A \times I = A \quad | \quad B \times I = B \quad \checkmark$$

It is monoid.

For inverse property, $a \times a^{-1} = a^{-1} \times a = e$
& $a^{-1} \in G$

$$\text{For } a = A, \quad A \times A^{-1} = A$$

$$\text{For } a = B, \quad B \times B^{-1} = A$$

$$\text{For } a = C, \quad C \times C^{-1} = A$$

$$\text{For } a = D, \quad D \times D^{-1} = A$$

It is a group.